

Answer ALL SIX questions from Section A and ANY TWO questions from Section B

The numbers in square brackets in the right-hand margin indicate the provisional allocation of maximum marks per sub-section of a question.

You may find the following constants and theorems useful.
The three Pauli matrices are:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Perturbation Theory:

$$E_n^{(1)} = \langle \psi_n^{(0)} | \lambda V' | \psi_n^{(0)} \rangle.$$

$$E_n^{(2)} = \sum_{k \neq n} \frac{\left| \langle \psi_k^{(0)} | \lambda V' | \psi_n^{(0)} \rangle \right|^2}{\left(E_n^{(0)} - E_k^{(0)} \right)}.$$

Section A

[Part marks]

(Answer ALL SIX questions from this section)

1. If $\hat{A}$ represents a Hermitian operator, show that:
 - (a) Its eigenvalues are real. [2]
 - (b) Its eigenvectors are orthogonal; state any assumptions you make. [4]

2. $|u_n\rangle$ form a complete set of states. Using Dirac notation, express the following:
 - (a) That $|u_n\rangle$ are orthonormal. [2]
 - (b) That an arbitrary operator $\hat{A}$ is Hermitian. [2]
 - (c) The resolution of the identity or closure relationship. [2]

3. (a) Show that if two operators have the same set of eigenvectors, they must commute. [4]
 - (b) Describe *briefly* the difference between the uncertainty in measurements on two observables whose operators $\hat{A}$ and $\hat{B}$ (i) commute or (ii) do not commute. Give formulae to quantify your answers. [3]

4. The operators $\hat{S}_x$, $\hat{S}_y$ and $\hat{S}_z$ representing the three Cartesian components of the spin angular momentum of a spin- $\frac{1}{2}$ particle can be expressed in terms of the Pauli spin matrices σ_x , σ_y and σ_z as:

$$\hat{S}_x = \frac{1}{2}\hbar\sigma_x, \quad \hat{S}_y = \frac{1}{2}\hbar\sigma_y, \quad \hat{S}_z = \frac{1}{2}\hbar\sigma_z.$$

(The Pauli spin matrices are given above in the rubric.)

- (a) What are the eigenvalues and eigenvectors of $\hat{S}_z$? [2]
- (b) The particle is in the spin-up state (its spin points along the z -axis). In this state, measurements are made of the component $\hat{S}_n = \mathbf{n} \cdot \hat{\mathbf{S}}$ of its spin angular momentum along the unit vector $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$, where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors along the x -, y - and z -axes. What are the possible outcomes of these measurements? (A statement of the possible outcomes is sufficient; you do not need to give a mathematical derivation of your answer.) [2]
- Evaluate the expectation value of S_n in the given state? [4]

- 5 (a) The Hamiltonian $\hat{H}$ of a system has normalised eigenvectors $|\phi_n\rangle$ and energy eigenvalues E_n . The system is prepared in the state $|\psi\rangle$ given by:

$$|\psi\rangle = \frac{1}{2}|\phi_0\rangle - \frac{\sqrt{3}}{2}|\phi_1\rangle,$$

where $|\phi_0\rangle$ and $|\phi_1\rangle$ are the eigenvectors of the ground state and the first excited state, with energies E_0 and E_1 . What is the value of $\langle\psi|\psi\rangle$? [2]

- (b) In the state $|\psi\rangle$, a measurement of the energy is made.
- i. What are the possible outcomes of the measurement and the probabilities of each outcome? [2]
- ii. Obtain the expectation value of the energy in the state $|\psi\rangle$. [2]

6. (a) Show, using Dirac notation, how an operator $\hat{A}$ may be represented by a matrix, using a complete set of orthonormal states $|u_n\rangle$. [4]
- (b) Two quantum states $|\psi\rangle$ and $|\chi\rangle$ are represented using a basis of orthonormal states. Obtain an expression for $\langle\psi|\chi\rangle$ in terms of the expansion coefficients of each state, defining any terms you may use. [3]

Section B

(Answer ANY TWO questions from this Section)

[Part marks]

Note: only two Section B answers will be marked

7. Perturbation theory expressions for the first-order and second-order the correction to the energy, $E_n^{(0)}$, of a system perturbed by a potential $\lambda V'$ are given on the front cover. Explain the circumstances where these expressions are useful. [6]

A particle of mass m sits in a square well potential defined by $V(x) = 0$ for $-a \leq x \leq a$ and $V = \infty$ for $|x| > a$. Its energy levels are given by

$$E_n^{(0)} = \frac{\hbar^2 \pi^2 n^2}{8ma^2}$$

with $n = 1, 2, \dots$. The wavefunction of the $n = 1$ ground state is given by

$$|1\rangle = \cos\left(\frac{\pi x}{2a}\right).$$

Normalise this function; you can assume the relationship $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$. [3]

Consider the following two perturbations

- (a) The potential well is raised to V_0 for the entire range $-a \leq x \leq a$.
- (b) The potential well is raised to V_0 for the half range $-a \leq x \leq 0$ only.

In each case obtain the first-order correction to the ground state energy. [7]

By considering the second-order correction to the energy, or otherwise, but without doing explicit calculations, explain how you expect these expressions to change as higher-order terms are added. [7]

Hence, or otherwise, give a general expression for the energy of the n^{th} level as the case (a) where the bottom of the potential well is raised to V_0 for the entire range $-a \leq x \leq a$. [3]

Considering the half-range potential of case (b). If

$$V_0 = \alpha \frac{\hbar^2 \pi^2}{ma^2},$$

comment on the use of perturbation theory when (i) $\alpha = 0.005$, (ii) $\alpha = 0.05$ and (iii) $\alpha = 0.5$. [4]

8. $\hat{F}$ is a general angular momentum operator which obeys standard commutation relations and whose associated step-up and step-down operators can be defined as

$$\hat{F}_+ = \hat{F}_x + i\hat{F}_y$$

$$\hat{F}_- = \hat{F}_x - i\hat{F}_y$$

- (a) Show that

i. $[\hat{F}_-, \hat{F}_+] = -2\hbar\hat{F}_z.$ [3]

ii. $[\hat{F}_-, \hat{F}_z] = \hbar\hat{F}_-.$ [2]

- (b) $\hat{F}^2$ and $\hat{F}_z$ have simultaneous eigenvectors $|f, m_f\rangle$ with eigenvalues $f(f+1)\hbar^2$ and $m_f\hbar$, respectively. Show that for $-f+1 \geq m_f \geq f$, the state $\hat{F}_-|f, m_f\rangle$ is an eigenvector of $\hat{F}_z$ and obtain the corresponding eigenvalue. Explain what happens when $\hat{F}_-$ operates on eigenvector $|f, -f\rangle$. [4]

- (c) $\hat{L}$ and $\hat{S}$ are the angular momentum operators associated with the electron orbital motion and spin, respectively. Given that $\hat{J} = \hat{L} + \hat{S}$, show that [4]

$$\hat{J}^2 = \hat{L}^2 + \hat{S}^2 + \hat{L}_+\hat{S}_- + \hat{L}_-\hat{S}_+ + 2\hat{L}_z\hat{S}_z$$

- (d) Consider the state denoted $|l, m; s, m_s\rangle = |l, m\rangle|s, m_s\rangle$. Demonstrate that $|l, -l; s, -s\rangle$ is an eigenvector of $\hat{J}^2$. Find the corresponding eigenvalue. [5]

- (e) An operator $\hat{Q} = \alpha\hat{L}^2 + \beta\hat{S}_+\hat{L}_z$ acts on the basis $|l=1, m; s=\frac{1}{2}, m_s\rangle$.

- i. Explain why the matrix representation of $\hat{Q}$ is a 6×6 matrix. [2]

- ii. Obtain this matrix.

You may assume that $\hat{S}_+|s, m_s\rangle = \hbar[s(s+1) - m_s(m_s+1)]^{\frac{1}{2}}|s, m_s+1\rangle$. [8]

- iii. Is $\hat{Q}$ Hermitian? [2]

9. The Hamiltonian for a one-dimensional harmonic oscillator is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where $\hat{x}$ and $\hat{p}$ are the position and momentum operators, m the particle mass and ω the angular frequency. The raising and lowering operators, $\hat{a}_+$ and $\hat{a}_-$, respectively are defined by

$$\hat{a}_+ = \left(\frac{1}{2\hbar m\omega}\right)^{1/2} (m\omega\hat{x} - i\hat{p}); \quad \hat{a}_- = \left(\frac{1}{2\hbar m\omega}\right)^{1/2} (m\omega\hat{x} + i\hat{p}).$$

They have the property $\hat{a}_+|n\rangle = \sqrt{n+1}|n+1\rangle$ and $\hat{a}_-|n\rangle = \sqrt{n}|n-1\rangle$, where $|n\rangle$ is the n^{th} eigenstate.

(a) Show that the Hamiltonian can be expressed as

$$\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right)$$

with $\hat{N} = \hat{a}_+\hat{a}_-$, the occupation number operator. [3]

(b) Express the position operator $\hat{x}$ in terms of $\hat{a}_+$ and $\hat{a}_-$. [3]

(c) Derive the value of the commutator $[\hat{a}_-, \hat{a}_+]$. [2]

(d) If $|\lambda\rangle$ is a normalized eigenstate of $\hat{N}$ with eigenvalue λ , i.e. $\hat{N}|\lambda\rangle = \lambda|\lambda\rangle$, show that

i. the eigenvalues $\lambda \geq 0$. [2]

ii. $\hat{a}_-|\lambda\rangle$ is an eigenstate of $\hat{N}$, with eigenvalue $\lambda - 1$. [4]

iii. Hence show that the eigenvalues of $\hat{N}$ are integers $\lambda = 0, 1, 2, \dots$ [3]

(e) The leading anharmonic correction to a harmonic oscillator system takes the form $-\lambda\hat{x}^3$, so that the Hamiltonian becomes

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 - \lambda\hat{x}^3.$$

By considering successive applications of the position operator $\hat{x}$ to the ground state $|0\rangle$, or otherwise, show that

$$\hat{x}^3|0\rangle = \left(\frac{\hbar}{2m\omega}\right)^{3/2} [\sqrt{6}|3\rangle + 3|1\rangle]$$

(f) Determine the ground state energy to second-order in λ . [7]

10. (a) The variational principle states that

$$\frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0$$

where $\hat{H}$ is a Hamiltonian for a system, E_0 is the energy of the lowest state of that system and ψ is any wavefunction for the system. Prove this statement.

[6]

- (b) The Hamiltonian for a system with a δ -function potential can be written

$$\hat{H} = \hat{T} + V$$

where

$$\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \quad \text{and} \quad V = -\alpha \delta(x)$$

A suitable trial wavefunction for this Hamiltonian can be written

$$|\psi\rangle \equiv \psi(x) = A \exp(-bx^2).$$

Evaluate:

- i. $\langle \psi | \psi \rangle$.

Explain how this can be used to determine the value of A .

[5]

- ii. $\langle T \rangle = \langle \psi | \hat{T} | \psi \rangle$

[4]

- iii. $\langle V \rangle = \langle \psi | V | \psi \rangle$

[2]

- iv. Hence show that an upper limit for the lowest state of this system, E_0 , in terms of the parameter b is given by $\frac{b\hbar^2}{2m} - \alpha \sqrt{\frac{2b}{\pi}}$.

[3]

- v. Obtain the lowest value possible for E_0 for this trial function.

[5]

- vi. The exact value for E_0 is $-\frac{m\alpha^2}{\pi\hbar^2}$. Comment on how your value compares with this. Suggest how an improved value might be obtained.

[5]

The following integrals may be assumed:

$$\int_{-\infty}^{+\infty} \exp(-ax^2) dx = \sqrt{\frac{\pi}{a}}, \quad \int_{-\infty}^{+\infty} x^2 \exp(-ax^2) dx = \frac{1}{2} \sqrt{\frac{\pi}{a^3}}$$